

Elementary Number Theory

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Source Code & Contributions

GitHub: <https://github.com/sihyeon8240/math>

Feel free to open issues or pull requests for corrections.

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Contents

Preface	v
1 Preliminaries	1
1.1 Mathematical Induction	1
1.2 The Binomial Theorem	1
2 Divisibility Theory in the Integers	2
2.1 Early Number Theory	2
2.2 The Division Algorithm	2
2.3 The Greatest Common Divisor	4
2.4 The Euclidean Algorithm	15
2.5 The Diophantine Equation $ax+by=c$	15
3 Primes and Their Distribution	16
3.1 The Fundamental Theorem of Arithmetic	16
3.2 The Sieve of Eratosthenes	16
3.3 The Goldbach Conjecture	16
4 The Theory of Congruences	17
4.1 Carl Friedrich Gauss	17
4.2 Basic Properties of Congruence	17
4.3 Binary and Decimal Representations of Integers	17
4.4 Linear Congruences and the Chinese Remainder Theorem	17
5 Fermat's Theorem	18
5.1 Pierre de Fermat	18
5.2 Fermat's Little Theorem and Pseudoprimes	18
5.3 Wilson's Theorem	18
5.4 The Fermat-Kraitichik Factorization Method	18
6 Number-Theoretic Functions	19
6.1 The Sum and Number of Divisors	19
6.2 The Möbius Inversion Formula	19

6.3	The Greatest Integer Function	19
6.4	An Application to the Calendar	19
7	Euler's Generalization of Fermat's Theorem	20
7.1	Leonhard Euler	20
7.2	Euler's Phi-Function	20
7.3	Euler's Theorem	20
7.4	Some Properties of the Phi-Function	20
8	Primitive Roots and Indices	21
8.1	The Order of an Integer Modulo n	21
8.2	Primitive Roots for Primes	21
8.3	Composite Numbers Having Primitive Roots	21
8.4	The Theory of Indices	21
9	The Quadratic Reciprocity Law	22
9.1	Euler's Criterion	22
9.2	The Legendre Symbol and Its Properties	22
9.3	Quadratic Reciprocity	22
9.4	Quadratic Congruences with Composite Moduli	22
10	Introduction to Cryptography	23
10.1	From Caesar Cipher to Public Key Cryptography	23
10.2	The Knapsack Cryptosystem	23
10.3	An Application of Primitive Roots to Cryptography	23
11	Numbers of Special Form	24
11.1	Marin Mersenne	24
11.2	Perfect Numbers	24
11.3	Mersenne Primes and Amicable Numbers	24
11.4	Fermat Numbers	24
12	Certain Nonlinear Diophantine Equations	25
12.1	The Equation $x^2 + y^2 = z^2$	25
12.2	Fermat's Last Theorem	25
13	Representation of Integers as Sums of Squares	26
13.1	Joseph Louis Lagrange	26
13.2	Sums of Two Squares	26
13.3	Sums of More Than Two Squares	26

14 Fibonacci Numbers	27
14.1 Fibonacci	27
14.2 The Fibonacci Sequence	27
14.3 Certain Identities Involving Fibonacci Numbers	27
15 Continued Fractions	28
15.1 Srinivasa Ramanujan	28
15.2 Finite Continued Fractions	28
15.3 Infinite Continued Fractions	28
15.4 Farey Fractions	28
15.5 Pell’s Equation	28
16 Some Modern Developments	29
16.1 Hardy, Dickson, and Erdős	29
16.2 Primality Testing and Factorization	29
16.3 An Application to Factoring: Remote Coin Flipping	29
16.4 The Prime Number Theorem and Zeta Function	29

Preface

This textbook uses Lean 4 to formalize selected mathematical statements and proofs.

- **Tools.** Lean is an interactive theorem prover; Mathlib is a library of formalized mathematics. This edition uses Lean 4.33.1 and Mathlib v4.33.1 (revision [0df444a360ea](#)).
- **Reading the proofs.** Each paired statement or proof step gives a mathematical explanation followed by its Lean code. The explanations can be read independently; code fragments are read in order within the complete declaration.
- **Verification.** Lean checks formal proofs under their stated assumptions. Verification covers only statements linked to checked declarations. Agreement between the explanations, displayed code, and checked proofs is reviewed separately.
- **Prerequisites.** Starting assumptions and accepted external results are specific to this textbook.

Chapter 1

Preliminaries

1.1 Mathematical Induction

1.2 The Binomial Theorem

Chapter 2

Divisibility Theory in the Integers

2.1 Early Number Theory

2.2 The Division Algorithm

Theorem 2.2.1.

Let $a \in \mathbb{Z}$ and $b \in \mathbb{N}^+$. There is a unique pair $(q, r) \in \mathbb{Z} \times \mathbb{N}$ such that

$$a = qb + r, \quad r < b.$$

We call q the **quotient** and r the **remainder**.

```
theorem existsUnique_quotient_remainder (a : ℤ) (b : ℕ+) :  
  ∃! qr : ℤ × ℕ, a = qr.1 * b + qr.2 ∧ qr.2 < b
```

Proof.

Since $b \geq 1$,

$$a - (-|a|)b = a + |a|b \geq a + |a| \geq 0.$$

```
classical  
have hb : (1 : ℤ) ≤ (b : ℤ) := by exact_mod_cast b.pos  
have hnonneg : 0 ≤ a - (-|a|) * (b : ℤ) := by  
  have ha : -|a| ≤ a := neg_abs_le a  
  have habs : 0 ≤ |a| := abs_nonneg a  
  nlinarith
```

Define

$$S := \{n \in \mathbb{N} : a - qb = n \text{ for some } q \in \mathbb{Z}\}.$$

By the preceding inequality, $S \neq \emptyset$. Let $r = \min S$ and choose $q \in \mathbb{Z}$ with $a - qb = r$.

```

let h :  $\exists n : \mathbb{N}, \exists q : \mathbb{Z}, a - q * b = (n : \mathbb{Z}) :=$ 
  ( $(a - (-|a|) * b).toNat, -|a|, (Int.toNat_of_nonneg hnonneg).symm$ )
obtain ⟨q, hq⟩ := Nat.find_spec h
let r :  $\mathbb{N} :=$  Nat.find h
have hr0 :  $(0 : \mathbb{Z}) \leq r :=$  Int.natCast_nonneg _

```

By minimality, for $s \in \mathbb{Z}$,

$$0 \leq s, \quad s = a - xb \ (x \in \mathbb{Z}) \quad \implies \quad r \leq s.$$

```

have hminimal :  $\forall s : \mathbb{Z}, 0 \leq s \rightarrow (\exists x : \mathbb{Z}, a - x * b = s) \rightarrow (r : \mathbb{Z}) \leq s$ 
   $\hookrightarrow :=$  by
    intro s hs ⟨x, hx⟩
    have hn : Nat.find h  $\leq$  s.toNat :=
      Nat.find_min' h ⟨x, hx.trans (Int.toNat_of_nonneg hs).symm⟩
    have hc : (Nat.find h :  $\mathbb{Z}$ )  $\leq$  (s.toNat :  $\mathbb{Z}$ ) := Int.ofNat_le.mpr hn
    rwa [Int.toNat_of_nonneg hs] at hc

```

If $r \geq b$, then

$$0 \leq r - b = a - (q + 1)b < r,$$

contradicting minimality. Hence $r < b$.

```

have hrb_int :  $(r : \mathbb{Z}) < b :=$  by
  by_contra hnot
  have hs :  $0 \leq (r : \mathbb{Z}) - b :=$  by linarith
  have hsmaller := hminimal ((r :  $\mathbb{Z}$ ) - b) hs ⟨q + 1, by dsimp [r];
     $\hookrightarrow$  nlinarith [hq]⟩
  linarith

```

Thus $a = qb + r$ and $0 \leq r < b$, proving existence.

```

have ha :  $a = q * b + r :=$  by dsimp [r]; linarith [hq]
have hrb :  $r < (b : \mathbb{N}) :=$  by exact_mod_cast hrb_int
refine ⟨(q, r), ⟨ha, hrb⟩, ?_⟩

```

Suppose another pair $(q', r') \in \mathbb{Z} \times \mathbb{N}$ satisfies $a = q'b + r'$ and $0 \leq r' < b$.

```
rintro (q', r') (ha', hr'b)
change a = q' * b + r' at ha'
have hr'0 : (0 : ℤ) ≤ r' := Int.natCast_nonneg _
have hr'b_int : (r' : ℤ) < b := by exact_mod_cast hr'b
```

Since $q, q' \in \mathbb{Z}$,

$$\begin{aligned} q' < q &\implies q - q' \geq 1 \implies r' = r + (q - q')b \geq b, \\ q' > q &\implies q' - q \geq 1 \implies r = r' + (q' - q)b \geq b. \end{aligned}$$

Both contradict the remainder bounds, so $q' = q$.

```
have hqq : q' = q := by
  rcases lt_trichotomy q' q with hlt | heq | hgt
  · have hstep : q' + 1 ≤ q := hlt
    nlinarith
  · exact heq
  · have hstep : q + 1 ≤ q' := hgt
    nlinarith
```

Then $r' = a - q'b = a - qb = r$, so $(q', r') = (q, r)$, proving uniqueness.

```
have hrr_int : (r' : ℤ) = r := by rw [hqq] at ha'; linarith
have hrr : r' = r := by exact_mod_cast hrr_int
exact Prod.ext hqq hrr
```

□

2.3 The Greatest Common Divisor

For integers c, z , define $c \mid z$ to mean that $z = ck$ for some $k \in \mathbb{Z}$. We use this existential definition directly, including when $c = 0$. The Lean declarations below belong to `ElementaryNumberTheory.Chapter02`.

Theorem 2.3.1.

Suppose $a, b \in \mathbb{Z}$ are not both zero. There is a positive natural number d which divides both a and b , is an integer linear combination of them, and is divisible by every common divisor.

```
theorem exists_positive_common_divisor (a b : ℤ) (hab : a ≠ 0 ∨ b ≠ 0) :
  ∃ d : ℕ, 0 < d ∧ (d : ℤ) ∣ a ∧ (d : ℤ) ∣ b ∧
    (∃ x y : ℤ, (d : ℤ) = a * x + b * y) ∧
    (∀ c : ℤ, c ∣ a → c ∣ b → c ∣ (d : ℤ))
```

Proof.

Since a, b are not both zero, $a^2 + b^2 > 0$. Thus the set of positive natural numbers of the form $ax + by$, with $x, y \in \mathbb{Z}$, is nonempty.

```
classical

have hpos : 0 < a * a + b * b := by
  rcases hab with ha | hb
  · nlinarith [sq_pos_of_ne_zero ha, sq_nonneg b]
  · nlinarith [sq_nonneg a, sq_pos_of_ne_zero hb]

have hS : ∃ n : ℕ, 0 < n ∧ ∃ x y : ℤ, a * x + b * y = (n : ℤ) := by
  refine ((a * a + b * b).toNat, ?_, a, b, ?_)
  · omega
  · exact (Int.toNat_of_nonneg (le_of_lt hpos)).symm
```

Let d be its least element and choose $x, y \in \mathbb{Z}$ with $ax + by = d$. Every positive natural number represented by such a combination is at least d .

```
let d : ℕ := Nat.find hS

obtain (hdpos, x, y, hxy) := Nat.find_spec hS

have hminimal (n : ℕ) (hn : 0 < n)
  (hlinear : ∃ u v : ℤ, a * u + b * v = (n : ℤ)) : d ≤ n :=
  Nat.find_min' hS (hn, hlinear)
```

Suppose $u, v \in \mathbb{Z}$. Apply the division algorithm to $au + bv$ and the positive divisor d , obtaining $au + bv = qd + r$ with $0 \leq r < d$.

```

have hddiv (u v : ℤ) : (d : ℤ) | a * u + b * v := by
  obtain ⟨(q, r), ⟨heq, hr⟩, _⟩ :=
    existsUnique_quotient_remainder (a * u + b * v) (d, hnpos)

  change a * u + b * v = q * (d : ℤ) + (r : ℤ) at heq
  change r < d at hr

```

The remainder is again a linear combination: $r = a(u - qx) + b(v - qy)$.

```

have hrlinear : a * (u - q * x) + b * (v - q * y) = (r : ℤ) := by
  calc
    a * (u - q * x) + b * (v - q * y) =
      (a * u + b * v) - q * (a * x + b * y) := by ring
    _ = r := by
      rw [hxy]
      linarith [heq]

```

If $r \neq 0$, then $r > 0$ and minimality gives $d \leq r$, contradicting $r < d$. Hence $r = 0$.

```

have hrzero : r = 0 := by
  by_contra hne

  have hle := hminimal r (Nat.pos_of_ne_zero hne)
    ⟨u - q * x, v - q * y, hrlinear⟩

  omega

```

Therefore $d \mid au + bv$. Taking $(u, v) = (1, 0)$ and $(0, 1)$ gives $d \mid a$ and $d \mid b$.

```

refine ⟨q, ?_⟩
rw [hrzero] at heq
simp only [Nat.cast_zero, add_zero, mul_comm] using heq

have hda : (d : ℤ) | a := by
  simp only [mul_one, mul_zero, add_zero] using hddiv 1 0

have hdb : (d : ℤ) | b := by
  simp only [mul_one, mul_zero, zero_add] using hddiv 0 1

```

Suppose $c \mid a$ and $c \mid b$. Choose $u, v \in \mathbb{Z}$ with $a = cu$ and $b = cv$. Then $d = ax + by = c(ux + vy)$, so $c \mid d$. This proves all the asserted properties.

```

have hcommon (c : ℤ) (hca : c ∣ a) (hcb : c ∣ b) : c ∣ (d : ℤ) := by
  obtain ⟨u, hu⟩ := hca
  obtain ⟨v, hv⟩ := hcb

  refine ⟨u * x + v * y, ?_⟩
  rw [← hxy, hu, hv]
  ring

exact ⟨d, hdpos, hda, hdb, ⟨x, y, hxy.symm⟩, hcommon⟩

```

□

Definition 2.3.2.

For $a, b \in \mathbb{Z}$ not both zero, define $\gcd(a, b)$ to be the positive common divisor constructed above from the least positive linear combination. Define $\gcd(0, 0) = 0$.

```

noncomputable def gcd (a b : ℤ) : ℕ :=
  if hab : a ≠ 0 ∨ b ≠ 0 then
    Classical.choose (exists_positive_common_divisor a b hab)
  else 0

```

Theorem 2.3.3.

For integers a, b not both zero, $\gcd(a, b)$ is positive, divides both inputs, is an integer linear combination of them, and is divisible by every common divisor.

```

theorem gcd_spec (a b : ℤ) (hab : a ≠ 0 ∨ b ≠ 0) :
  0 < gcd a b ∧ (gcd a b : ℤ) ∣ a ∧ (gcd a b : ℤ) ∣ b ∧
  (∃ x y : ℤ, (gcd a b : ℤ) = a * x + b * y) ∧
  (∀ c : ℤ, c ∣ a → c ∣ b → c ∣ (gcd a b : ℤ))

```

Proof.

The definition selects the number constructed above, together with its established properties.

```

rw [gcd, dif_pos hab]
exact Classical.choose_spec (exists_positive_common_divisor a b hab)

```

□

Theorem 2.3.4.

Suppose $a, b \in \mathbb{Z}$ are not both zero and $c \in \mathbb{Z}$ divides both. Then $c \leq \gcd(a, b)$.

```
theorem common_divisor_le_gcd (a b c :  $\mathbb{Z}$ ) (hab : a  $\neq$  0  $\vee$  b  $\neq$  0)
  (hca : c | a) (hcb : c | b) : c  $\leq$  (gcd a b :  $\mathbb{Z}$ )
```

Proof.

Write $d = \gcd(a, b) > 0$. Since $c \mid d$, there is $k \in \mathbb{Z}$ with $d = ck$. If $c > 0$, positivity of d forces $k > 0$, hence $k \geq 1$ and $d \geq c$. If $c \leq 0$, then $c < d$ immediately.

```
obtain ⟨hpos, _, _, _, hcommon⟩ := gcd_spec a b hab
have hdpos : (0 :  $\mathbb{Z}$ ) < gcd a b := by exact_mod_cast hpos
obtain ⟨k, hk⟩ := hcommon c hca hcb
by_cases hc : 0 < c
· have hkpos : 0 < k := by nlinarith
  have hkone : 1  $\leq$  k := hkpos
  nlinarith
· linarith
```

□

Theorem 2.3.5.

The zero-pair convention is $\gcd(0, 0) = 0$.

```
theorem gcd_zero_zero : gcd 0 0 = 0
```

Proof.

Neither input is nonzero, so the definition assigns the value zero. This is a convention; the common divisors of zero and zero have no greatest integer.

```
unfold gcd
rw [dif_neg (by rintro (h | h); exact h rfl; exact h rfl)]
```

□

Theorem 2.3.6.

Suppose $a, b \in \mathbb{Z}$ are not both zero and $d \in \mathbb{N}$ divides both. If every integer common divisor is at most d , then $\gcd(a, b) = d$.

```
theorem gcd_eq_of_greatest (a b :  $\mathbb{Z}$ ) (hab :  $a \neq 0 \vee b \neq 0$ ) (d :  $\mathbb{N}$ )
  (hda : (d :  $\mathbb{Z}$ ) | a) (hdb : (d :  $\mathbb{Z}$ ) | b)
  (hgreatest :  $\forall c : \mathbb{Z}, c | a \rightarrow c | b \rightarrow c \leq (d : \mathbb{Z})$ ) : gcd a b = d
```

Proof.

Since $\gcd(a, b)$ is a common divisor, the hypothesis gives $\gcd(a, b) \leq d$. The preceding bound applied to d gives $d \leq \gcd(a, b)$. Antisymmetry proves equality.

```
have hspec := gcd_spec a b hab
have hle := hgreatest (gcd a b) hspec.2.1 hspec.2.2.1
have hge := common_divisor_le_gcd a b d hab hda hdb
exact_mod_cast (le_antisymm hle hge)
```

□

Theorem 2.3.7.

Suppose $a, b \in \mathbb{Z}$ are not both zero. There are integers x, y such that $\gcd(a, b) = ax + by$.

```
theorem exists_gcd_eq_mul_add_mul (a b :  $\mathbb{Z}$ ) (hab :  $a \neq 0 \vee b \neq 0$ ) :
   $\exists x y : \mathbb{Z}, (\gcd a b : \mathbb{Z}) = a * x + b * y$ 
```

Proof.

The linear-combination representation is one of the properties established when constructing the gcd.

```
exact (gcd_spec a b hab).2.2.2.1
```

□

Theorem 2.3.8.

Suppose $a, b \in \mathbb{Z}$ are not both zero. The set of integer linear combinations $ax + by$ is exactly the set of integer multiples of $\gcd(a, b)$.

```
theorem linear_combinations_eq_gcd_multiples (a b : ℤ) (hab : a ≠ 0 ∨ b
↪ ≠ 0) :
  {z : ℤ | ∃ x y : ℤ, z = a * x + b * y} =
  {z : ℤ | ∃ n : ℤ, z = n * (gcd a b : ℤ)}
```

Proof.

Write $d = \gcd(a, b) = ax_0 + by_0$. Since d divides both inputs, choose $u, v \in \mathbb{Z}$ with $a = du$ and $b = dv$. We prove both set inclusions.

```
obtain (x₀, y₀, hbezout) := exists_gcd_eq_mul_add_mul a b hab
have hcommon := (gcd_spec a b hab).2
obtain (u, hu) := hcommon.1
obtain (v, hv) := hcommon.2.1

apply Set.ext
intro z
constructor
```

If $z = ax + by$, then $z = (ux + vy)d$, so z is an integer multiple of d .

```
· rintro (x, y, hz)

refine (u * x + v * y, ?_)
calc
  z = a * x + b * y := hz
  _ = ((gcd a b : ℤ) * u) * x + ((gcd a b : ℤ) * v) * y := by
    rw [← hu, ← hv]
  _ = (u * x + v * y) * (gcd a b : ℤ) := by ring
```

Conversely, if $z = nd$, then $z = a(nx_0) + b(ny_0)$, an integer linear combination of a, b .

```
· rintro (n, hz)

refine (n * x₀, n * y₀, ?_)
rw [hz, hbezout]
ring
```

□

Theorem 2.3.9.

Let $a, b \in \mathbb{Z}$ be not both zero. They are relatively prime, meaning $\gcd(a, b) = 1$, if and only if there are integers x, y with $1 = ax + by$.

```
theorem gcd_eq_one_iff_exists_mul_add_mul (a b : ℤ) (hab : a ≠ 0 ∨ b ≠
↪ 0) :
  gcd a b = 1 ↔ ∃ x y : ℤ, 1 = a * x + b * y
```

Proof.

For the forward implication, substitute $\gcd(a, b) = 1$ in the previously proved Bezout identity.

```
constructor
· intro h
  obtain ⟨x, y, hxy⟩ := exists_gcd_eq_mul_add_mul a b hab
  exact ⟨x, y, by simp only [h, Nat.cast_one] using hxy⟩
```

Conversely, 1 divides both inputs. If c is a common divisor, write $a = cu$ and $b = cv$. Then $1 = c(ux + vy)$. For $c > 0$ the second factor is a positive integer, hence at least one, so $c \leq 1$; for $c \leq 0$ the same bound is immediate. The order characterization of the gcd gives the result.

```
· rintro ⟨x, y, hxy⟩
  apply gcd_eq_of_greatest a b hab 1
  · exact ⟨a, by ring⟩
  · exact ⟨b, by ring⟩
  · intro c hca hcb
    obtain ⟨u, hu⟩ := hca
    obtain ⟨v, hv⟩ := hcb
    have hone : 1 = c * (u * x + v * y) := by
      calc
        1 = a * x + b * y := hxy
        _ = c * (u * x + v * y) := by rw [hu, hv]; ring
    by_cases hc : 0 < c
    · have hk : 0 < u * x + v * y := by nlinarith
      have hkone : 1 ≤ u * x + v * y := hk
      norm_num only [Nat.cast_one]
      nlinarith
    · norm_num only [Nat.cast_one]
      omega
```

□

Corollary 2.3.10.

For integers a, b not both zero, set $d = \gcd(a, b)$. Then $\gcd(a/d, b/d) = 1$, where the quotients are integers.

```
theorem gcd_div_gcd_eq_one (a b : ℤ) (hab : a ≠ 0 ∨ b ≠ 0) :
  gcd (a / (gcd a b : ℤ)) (b / (gcd a b : ℤ)) = 1
```

Proof.

Since the positive gcd divides both inputs, write $a = du$ and $b = dv$. The integer quotients are exactly u and v , since $d \neq 0$.

```
obtain ⟨hdpos, ⟨u, hu⟩, ⟨v, hv⟩, ⟨x, y, hxy⟩, _⟩ := gcd_spec a b hab
generalize hg : gcd a b = d at *
have hdne : (d : ℤ) ≠ 0 := by exact_mod_cast (ne_of_gt hdpos)
have ha : a / (d : ℤ) = u := by rw [hu]; exact Int.mul_ediv_cancel_left
  ↪ u hdne
have hb : b / (d : ℤ) = v := by rw [hv]; exact Int.mul_ediv_cancel_left
  ↪ v hdne
```

Choose Bezout coefficients with $d = ax + by$. Substitution gives $d = d(ux + vy)$, and cancellation of the nonzero integer d yields $1 = ux + vy$.

```
have hone : (1 : ℤ) = u * x + v * y := by
  apply mul_left_cancel₀ hdne
calc
  (d : ℤ) * 1 = a * x + b * y := by simp using hxy
  _ = (d : ℤ) * (u * x + v * y) := by rw [hu, hv]; ring
```

The identity rules out $u = v = 0$. Apply the preceding characterization to conclude that the two quotients are relatively prime.

```
have huv : u ≠ 0 ∨ v ≠ 0 := by
  by_contra h
  push Not at h
obtain ⟨rfl, rfl⟩ := h
  norm_num at hone
rw [ha, hb]
exact (gcd_eq_one_iff_exists_mul_add_mul u v huv).2 ⟨x, y, hone⟩
```

□

Corollary 2.3.11.

Let $a, b, c \in \mathbb{Z}$. If $\gcd(a, b) = 1$, $a \mid c$, and $b \mid c$, then $ab \mid c$.

```
theorem mul_dvd_of_coprime (a b c : ℤ) (hab : gcd a b = 1)
  (hac : a ∣ c) (hbc : b ∣ c) : a * b ∣ c
```

Proof.

The inputs cannot both be zero because $\gcd(0, 0) = 0$. Thus the coprimality characterization supplies integers x, y with $1 = ax + by$.

```
have hnz : a ≠ 0 ∨ b ≠ 0 := by
  by_contra h
  push Not at h
  obtain ⟨rfl, rfl⟩ := h
  rw [gcd_zero_zero] at hab
  contradiction
obtain ⟨x, y, hxy⟩ := (gcd_eq_one_iff_exists_mul_add_mul a b hnz).1 hab
```

Write $c = ar = bs$. Multiplying the Bezout identity by c and using these two expressions gives $c = acx + bcy = ab(sx + ry)$. The integer $sx + ry$ is the required divisibility witness.

```
obtain ⟨r, hr⟩ := hac
obtain ⟨s, hs⟩ := hbc
refine ⟨s * x + r * y, ?_⟩
calc
  c = c * (a * x + b * y) := by rw [← hxy]; ring
  _ = a * c * x + b * c * y := by ring
  _ = a * (b * s) * x + b * (a * r) * y := by nth_rw 1 [hs]; rw [hr]
  _ = (a * b) * (s * x + r * y) := by ring
```

□

Theorem 2.3.12.

Let $a, b, c \in \mathbb{Z}$ with $\gcd(a, b) = 1$. If $a \mid bc$, then $a \mid c$ (Euclid's lemma).

```
theorem dvd_of_dvd_mul_of_coprime (a b c : ℤ) (hab : gcd a b = 1)
  (hdiv : a ∣ b * c) : a ∣ c
```

Proof.

As before, coprimality excludes the zero pair, so choose integers x, y with $1 = ax + by$.

```

have hnz : a ≠ 0 ∨ b ≠ 0 := by
  by_contra h
  push Not at h
  obtain ⟨rfl, rfl⟩ := h
  rw [gcd_zero_zero] at hab
  contradiction
obtain ⟨x, y, hxy⟩ := (gcd_eq_one_iff_exists_mul_add_mul a b hnz).1 hab

```

Write $bc = ak$. Multiplication by c gives $c = acx + bcy = a(cx + ky)$, which exhibits an integer witness for $a \mid c$.

```

obtain ⟨k, hk⟩ := hdiv
refine ⟨c * x + k * y, ?_⟩
calc
  c = (a * x + b * y) * c := by rw [← hxy]; ring
  _ = a * (c * x) + (b * c) * y := by ring
  _ = a * (c * x + k * y) := by rw [hk]; ring

```

□

Theorem 2.3.13.

Let $a, b \in \mathbb{Z}$ be not both zero and let $d \in \mathbb{Z}$ be positive. Then $d = \gcd(a, b)$ if and only if d divides both inputs and every integer common divisor of a, b divides d .

```

theorem eq_gcd_iff_common_divisor (a b d : ℤ) (hab : a ≠ 0 ∨ b ≠ 0)
  (hd : 0 < d) :
  d = (gcd a b : ℤ) ↔
  d ∣ a ∧ d ∣ b ∧ (∀ c : ℤ, c ∣ a → c ∣ b → c ∣ d)

```

Proof.

If $d = \gcd(a, b)$, all three divisibility properties follow from the construction of the gcd.

```

constructor
· intro heq
  subst d
  obtain ⟨_, hda, hdb, _, hcommon⟩ := gcd_spec a b hab
  exact ⟨hda, hdb, hcommon⟩

```

Conversely, let $g = \gcd(a, b) > 0$. Because d is a common divisor, $d \leq g$. Since g divides both inputs, the last hypothesis gives $d = gk$. Positivity forces the integer k to be at least one, so $g \leq d$. The two bounds imply $d = g$.

```

• rintro (hda, hdb, hcommon)
  have hle := common_divisor_le_gcd a b d hab hda hdb
  obtain (hgpos, hga, hgb, _, _) := gcd_spec a b hab
  obtain (k, hk) := hcommon (gcd a b) hga hgb
  have hgpos' : (0 : ℤ) < gcd a b := by exact_mod_cast hgpos
  have hkpos : 0 < k := by nlinarith
  have hkone : 1 ≤ k := hkpos
  have hge : (gcd a b : ℤ) ≤ d := by nlinarith
  exact le_antisymm hle hge

```

□

2.4 The Euclidean Algorithm

2.5 The Diophantine Equation $ax+by=c$

Chapter 3

Primes and Their Distribution

3.1 The Fundamental Theorem of Arithmetic

3.2 The Sieve of Eratosthenes

3.3 The Goldbach Conjecture

Chapter 4

The Theory of Congruences

4.1 Carl Friedrich Gauss

4.2 Basic Properties of Congruence

4.3 Binary and Decimal Representations of Integers

4.4 Linear Congruences and the Chinese Remainder Theorem

Chapter 5

Fermat's Theorem

5.1 Pierre de Fermat

5.2 Fermat's Little Theorem and Pseudoprimes

5.3 Wilson's Theorem

5.4 The Fermat-Kraitchik Factorization Method

Chapter 6

Number-Theoretic Functions

6.1 The Sum and Number of Divisors

6.2 The Möbius Inversion Formula

6.3 The Greatest Integer Function

6.4 An Application to the Calendar

Chapter 7

Euler's Generalization of Fermat's Theorem

7.1 Leonhard Euler

7.2 Euler's Phi-Function

7.3 Euler's Theorem

7.4 Some Properties of the Phi-Function

Chapter 8

Primitive Roots and Indices

8.1 The Order of an Integer Modulo n

8.2 Primitive Roots for Primes

8.3 Composite Numbers Having Primitive Roots

8.4 The Theory of Indices

Chapter 9

The Quadratic Reciprocity Law

9.1 Euler's Criterion

9.2 The Legendre Symbol and Its Properties

9.3 Quadratic Reciprocity

9.4 Quadratic Congruences with Composite Moduli

Chapter 10

Introduction to Cryptography

10.1 From Caesar Cipher to Public Key Cryptography

10.2 The Knapsack Cryptosystem

10.3 An Application of Primitive Roots to Cryptography

Chapter 11

Numbers of Special Form

11.1 Marin Mersenne

11.2 Perfect Numbers

11.3 Mersenne Primes and Amicable Numbers

11.4 Fermat Numbers

Chapter 12

Certain Nonlinear Diophantine Equations

12.1 The Equation $x^2 + y^2 = z^2$

12.2 Fermat's Last Theorem

Chapter 13

Representation of Integers as Sums of Squares

13.1 Joseph Louis Lagrange

13.2 Sums of Two Squares

13.3 Sums of More Than Two Squares

Chapter 14

Fibonacci Numbers

14.1 Fibonacci

14.2 The Fibonacci Sequence

14.3 Certain Identities Involving Fibonacci Numbers

Chapter 15

Continued Fractions

15.1 Srinivasa Ramanujan

15.2 Finite Continued Fractions

15.3 Infinite Continued Fractions

15.4 Farey Fractions

15.5 Pell's Equation

Chapter 16

Some Modern Developments

16.1 Hardy, Dickson, and Erdős

16.2 Primality Testing and Factorization

16.3 An Application to Factoring: Remote Coin Flipping

16.4 The Prime Number Theorem and Zeta Function

Bibliography

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